

Australian Standard[®]

Bridge design

Part 3: Foundations and soil-supporting structures



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- Association of Consulting Engineers Australia
 - Australasian Railway Association
 - Austroads
 - Bureau of Steel Manufacturers of Australia
 - Cement and Concrete Association of Australia
 - Institution of Engineers Australia
 - Queensland University of Technology
 - Steel Reinforcement Institute of Australia
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-

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Part 3: Foundations and soil-supporting structures

Originated as HB 77.3—1996.
Revised and redesignated as AS 5100.3—2004.
Reissued incorporating Amendment No. 1 (April 2010).

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PREFACE

This Standard was prepared by the Standards Australia Committee BD-090, Bridge Design to supersede HB 77.3—1996, *Australian Bridge Design Code*, Section 3: *Foundations*.

This Standard incorporates Amendment No. 1 (April 2010). The changes required by the Amendment are indicated in the text by a marginal bar and amendment number against the clause, note, table, figure or part thereof affected.

The AS 5100 series represents a revision of the 1996 HB 77 series, *Australian Bridge Design Code*, which contained a separate Railway Supplement to Sections 1 to 5, together with Section 6, *Steel and composite construction*, and Section 7, *Rating*. AS 5100 takes the requirements of the Railway Supplement and incorporates them into Parts 1 to 5 of the present series, to form integrated documents covering requirements for both road and rail bridges. In addition, technical material has been updated.

This Standard is also designated as AUSTROADS publication AP-G15.3/04.

The objectives of AS 5100 are to provide nationally acceptable requirements for—

- (a) the design of road, rail, pedestrian and bicycle-path bridges;
- (b) the specific application of concrete, steel and composite steel/concrete construction, which embody principles that may be applied to other materials in association with relevant Standards; and
- (c) the assessment of the load capacity of existing bridges.

These requirements are based on the principles of structural mechanics and knowledge of material properties, for both the conceptual and detailed design, to achieve acceptable probabilities that the bridge or associated structure being designed will not become unfit for use during its design life.

Whereas earlier editions of the *Australian Bridge Design Code* were essentially administered by the infrastructure owners and applied to their own inventory, an increasing number of bridges are being built under the design-construct-operate principle and being handed over to the relevant statutory authority after several years of operation. This Standard includes clauses intended to facilitate the specification to the designer of the functional requirements of the owner, to ensure the long-term performance and serviceability of the bridge and associated structure.

Significant differences between this Standard and HB 77.3 are the following:

- (i) *Foundation design principles* In recognition that geotechnical engineering design principles differ from structural engineering design principles, the design procedures have been extensively revised. Designers are required to use geotechnical engineering methods appropriate to the foundation problem at hand, together with appropriate characteristic values and factors, when deriving economical and safe solutions. It is further required that designers apply engineering judgement to the application of sound rational design methods outlined in texts, technical literature and other design codes to supplement the design requirements of this Standard.
- (ii) *Design procedures* Substructures have been classified as either foundations, where most of the loads on the substructure come from the bridge structure and loads on it, or as soil-supporting structures, where most of the applied loads are from earth pressure. Different design procedures are required for each. The loads and resistances for a soil-supporting structure will largely depend on the soil properties, whereas the loads for a foundation will not be as dependent on the soil properties.

- (iii) *Relevant Standard* The philosophy used for the design of earth-retaining structures in this Standard differs from that contained in AS 4678, *Earth-retaining structures*, which was prepared by Standards Australia Committee CE-032. It is considered that for bridges and road-related structures, where soil/structure interaction occurs and the loads are predominantly soil-imposed, the design method adopted is more realistic. However, AS 4678 contains much useful information that can be used to supplement the design of structures covered by this Standard.

In line with Standards Australia policy, the words ‘shall’ and ‘may’ are used consistently throughout this Standard to indicate respectively, a mandatory provision and an acceptable or permissible alternative.

Statements expressed in mandatory terms in Notes to Tables are deemed to be requirements of this Standard.

The terms ‘normative’ and ‘informative’ have been used in this Standard to define the application of the appendix to which they apply. A ‘normative’ appendix is an integral part of the Standard, whereas an ‘informative’ appendix is only for information and guidance.

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STANDARDS AUSTRALIA

Australian Standard Bridge design

Part 3: Foundations and soil-supporting structures

1 SCOPE

This Standard sets out minimum design requirements and procedures for the design in limit states format of foundations and soil-supporting structures for road, rail and pedestrian bridges, culverts not specifically covered by other Standards, and subways of conventional size and form.

Foundations include shallow footings, piles and anchorages.

Soil-supporting structures include retaining walls, abutments and buried structures.

The provisions also covers the design of foundations for road furniture, such as lighting poles and sign support structures and noise barriers.

The Standard does not cover the design of—

- (a) corrugated steel pipes and arches (see AS 1762, AS/NZS 2041 and AS 3703.2);
- (b) underground concrete drainage pipes (see AS 3725 and AS 4058); and
- (c) reinforced soil structures.

The requirements for structural design and detailing of concrete and steel are specified in AS 5100.5 and AS 5100.6; however, a number of specific structural design provisions that result from soil-structure interaction are covered by this Standard.

2 APPLICATION

For the design of foundations for overhead wiring structures for electrified railway lines, the requirements of the relevant authority shall apply.

The loads to be applied shall be those specified in AS 5100.2, together with earth pressure loads determined in accordance with this Standard.

The general design procedures to be adopted shall be as specified in this Standard. Unless specified otherwise by the relevant authority, the detailed methods and formulae to be used shall be those specified in the relevant Standard for the geotechnical or structural element. Where no Australian Standard exists covering the design of the geotechnical or structural element, rational design methods outlined in texts or other design Standards and technical literature shall be used, as approved by the relevant authority.

3 REFERENCED DOCUMENTS

The following Standards are referred to in this Standard:

AS	
1597	Precast reinforced concrete box culverts
1597.2	Part 2: Large culverts (from 1500 mm span and up to and including 4200 mm span and 4200 mm height)
1726	Geotechnical site investigations
1762	Helical lock-seam corrugated steel pipes—Design and installation

AS	
2159	Piling—Design and installation
3703	Long-span corrugated steel structures
3703.2	Part 2: Design and installation
3725	Loads on buried concrete pipes
4058	Precast concrete pipes (pressure and non-pressure)
5100	Bridge design
5100.1	Part 1: Scope and general principles
5100.2	Part 2: Design loads
5100.5	Part 5: Concrete
5100.6	Part 6: Steel and composite construction
5100.3 Supp 1	Bridge design—Foundations and soil-supporting structures— Commentary (Supplement to AS 5100.3—2003)
AS/NZS	
1554	Structural steel welding
1554.1	Part 1: Welding of steel structures
1554.3	Part 3: Welding of reinforcing steel
2041	Buried corrugated metal structures

4 DEFINITIONS

For the purpose of this Standard, the definitions below apply. Definitions peculiar to the particular Clause are also given in that Clause.

4.1 Bond length

That length at the end of a tendon within which provision is made for the load transfer to the surrounding rock.

4.2 Design values

The values of variables entered into the calculations.

4.3 Design working load

The long-term load that is required in the tendon.

4.4 Effective free length

The apparent length over which the tendon is assumed to extend elastically, as determined by stressing tests.

4.5 Free length

That length of a tendon between the anchorage assembly and the bond length (or transition length) that does not transfer any tendon load to the surrounding rock, concrete or other material through which the anchor passes.

4.6 Geotechnical engineer

A suitably qualified engineer with relevant geotechnical experience in charge of geotechnical investigation or design, or both.

4.7 Initial load

The initial load selected for proof load and acceptance tests.

4.8 Lift-off test

The test to determine the residual load in the tendon.

4.9 Lock-off load

The load equal to the design working load plus an allowance for loss of prestress.

4.10 Minimum breaking load

The minimum breaking load of the tendon.

4.11 Residual load

The load remaining in the tendon at any time after lock-off, usually measured by a lift-off test.

4.12 Test load

The maximum load to which a tendon is subjected in the short term for proof load and acceptance tests.

5 NOTATION

The symbols used in this Standard are listed in Table 5.

TABLE 5
NOTATION

Symbol	Description	Clause reference
A_t	cross-sectional area of tendon (in millimetres square) as determined by testing	Paragraph B2.11
E_{pr}	ultimate passive resistance of the soil in front of the footing	10.3.3.4
E_t	modulus of elasticity of steel tendon (megapascals) as determined by testing	Paragraph B2.11
F_{em}	moments, forces or loads in the foundation induced by lateral ground movements	8.2.2
F_{es}	compressive and tensile loads in the foundation, structure or its element caused by vertical ground movement	8.2.2
F_{nf}	negative friction loads on the foundation caused by consolidation of surrounding soil	8.2.2
H_{ug}	ultimate shear resistance at the base of the footing	8.3.3.4
L_{ef}	effective free length	Paragraph B2.11
L_{fr}	free length	Paragraph B2.11
L_v	bond length	Paragraph B2.11
R_u	ultimate strength	12.3.3
R_{ug}	ultimate geotechnical strength	7.3.1
R_{us}	ultimate structural strength	7.3.1
R_a	anchorage resistance	12.6.2
R_{ak}	characteristic anchorage strength	12.6.2
R_{am}	measured anchorage capacity	12.6.2
S_e	soil imposed action effects	7.3.2
S^*	design action effects	7.3.1
T	anchor load	Paragraph B2.11
T_A	initial load	Paragraph B2.11
T_D	design working load	Paragraph B2.11

(continued)

TABLE 5 (continued)

Symbol	Description	Clause reference
T_o	lock-off load	Paragraph B2.11
T_p	test load	Paragraph B2.11
T_R	residual load	Paragraph B2.11
T_{RC}	calculated residual load immediately after lock-off	Paragraph B2.11
T_u	minimum breaking load	Paragraph B2.11
δL	total extension of tendon relative to a datum	Paragraph B2.11
δL_e	elastic extension of tendon at each load stage	Paragraph B2.11
δL_r	calculated elastic extension of tendon under test load (T_p)	Paragraph B2.11
δL_{pl}	plastic or non-recoverable extension of tendon at each load stage	Paragraph B2.11
ϕ	strength reduction factor	12.3.3
ϕ_c	conversion factor	12.6.2
ϕ_g	geotechnical strength reduction factor	7.3.1
ϕ_n	importance category reduction factor	12.3.3
ϕ_s	structural strength reduction factor	7.3.1

6 SITE INVESTIGATION

6.1 General

A site investigation shall be carried out for all structures, to provide the necessary geotechnical information required for the design and construction of foundations and soil-supporting structures.

The investigation shall be carried out under the supervision of a geotechnical engineer unless approved otherwise by the relevant authority.

The site investigations shall be carried out in accordance with AS 1726.

Investigations may be one of the following:

- (a) *Preliminary investigation* An investigation conducted at the feasibility stage in order to assess alternative sites or routes, to prepare conceptual designs, to determine preliminary costings and to define constraints for the design.

The extent and coverage of the preliminary investigation shall be as required by the relevant authority, and may include—

- (i) field reconnaissance;
- (ii) topography;
- (iii) hydrology;
- (iv) geomorphology;
- (v) hydrogeology;
- (vi) examination of neighbouring structures and excavations;
- (vii) geological and geotechnical maps and records;
- (viii) previous site investigations and construction experience in the vicinity;
- (ix) aerial photographs;
- (x) maps;
- (xi) regional seismicity; or
- (xii) any other relevant information.

- (b) *Design investigation* Design investigation shall provide sufficient geotechnical information for the design and construction of the project.

The extent and coverage of the design investigation shall be as required by the relevant authority, and shall include the following:

- (i) Nature and size of the structure and its elements, including any special requirements.
- (ii) Conditions with regard to the surroundings of the structure, such as neighbouring structures, traffic, utilities, services and utilities, hazardous chemicals and the like.
- (iii) Ground conditions with particular regard to geological complexity.
- (iv) Ground water conditions.
- (v) Regional seismicity.
- (vi) Influence of the environment of the structure, such as hydrology, surface water, subsidence and the like.
- (vii) Aggressivity of soil and ground water with respect to materials used in the structure, e.g., acid sulphate soils.
- (viii) Scour effects.
- (ix) Working in the vicinity of electrified railway lines.
- (x) Other relevant factors.

6.2 Design investigations

The number of boreholes or other in situ tests, or both, depends on the proposed structure and the inferred uniformity of the subsoil conditions.

Unless otherwise specified by the relevant authority, the minimum number of boreholes shall be as follows:

- (a) *For bridge foundations* One per pier and abutment.
- (b) *For culverts, retaining walls and the like* One at each end, and for intermediate locations, one at not more than 30 m intervals.

NOTES:

- 1 This minimum level of investigation would only be satisfactory for sites with relatively uniform subsoil strata and easily defined foundation conditions.
- 2 Additional boreholes or test locations would be required where bridge approaches involve cuttings or embankments, in order to check that these earthworks would not cause vertical or lateral ground movements, or slope instability, which could adversely affect the bridge or associated structures.

Boreholes, pits or other in situ tests, as required, shall extend through any strata that may influence strength, stability or serviceability, or otherwise influence foundations or soil-supporting structures during or after construction.

The presence of ground water and its effects shall be investigated.

NOTE: Specific ground water effects may include—

- (a) the level and fluctuations of the permanent water table;
- (b) the inflow rates into excavations;
- (c) effects of dewatering on the water table and on adjacent structures;
- (d) the presence of and pressures associated with artesian and subartesian conditions; and
- (e) the potential aggressiveness of the ground water to buried concrete, steel and the like.

The results of a geotechnical investigation shall be compiled in a geotechnical report verified by a geotechnical engineer.

7 DESIGN REQUIREMENTS

7.1 Aim

The aim of the design of structures covered by this Standard is to provide a foundation or soil-supporting structure that is durable, stable and has adequate strength while serving its intended function and that also satisfies other relevant requirements, such as robustness, ease of construction, minimum disruption of normal operations during construction and minimal effects on adjacent existing structures accounting for effects of future works. Foundation behaviour shall be compatible with the superstructure so that both remain serviceable and can perform their intended functions.

NOTE: Worked examples to demonstrate the design process are given in AS 5100.3 Supp 1.

7.2 Design

The design of foundations or soil-supporting structures shall take into account, as appropriate, strength, stability, serviceability, durability and other relevant design requirements in accordance with this Standard.

7.3 Design for strength

7.3.1 General

Foundations and soil-supporting structures shall be designed for both structural and geotechnical strength as follows:

- (a) For foundations where the loads are imposed predominantly from or via the structure or loads applied to the structure, e.g., shallow footings, piles and anchorages, the strength shall be determined in accordance with Clause 7.3.2.
- (b) For soil-supporting structures where the loads are predominantly soil-imposed loads, e.g., abutments and buried structures, the strength shall be determined in accordance with Clause 7.3.3.

Where structures act as both foundations and soil-supporting structures, e.g., diaphragm walls supporting bridge abutments, such structures shall be designed to satisfy the requirements of both foundations and soil-supporting structures.

7.3.2 Foundations

Foundations shall be designed as follows:

- (a) The appropriate loads and other actions shall be determined in accordance with Clause 8.2.
- (b) The loads and action effects shall be factored and combined in accordance with Clause 8.3.2, to determine the design action loads (S^*) for strength for the foundation and its components for each appropriate load combination.
- (c) The ultimate geotechnical strength (R_{ug}) and the ultimate structural strength (R_{us}) shall be determined in accordance with Clause 10, 11 or 12, and AS 5100.5 or AS 5100.6, as appropriate, using unfactored characteristic values of material parameters.
- (d) The foundation and structural components shall be proportioned so that—

$$\phi_g R_{ug} \geq S^* \quad \dots 7.3.2(1)$$

$$\phi_s R_{us} \geq S^* \quad \dots 7.3.2(2)$$

where (ϕ_g) is a geotechnical strength reduction factor and (ϕ_s) is a structural strength reduction factor. ϕ_g shall be selected in accordance with Clause 7.3.5.

7.3.3 Soil-supporting structures

Soil-supporting structures shall be designed as follows:

- (a) The appropriate loads and other actions shall be determined in accordance with Clause 8.2.
- (b) The loads and action effects shall be combined in accordance with Clause 8.3.3, to determine the design loads for strength and stability.
- (c) An appropriate engineering analysis shall be carried out with all loads and load combinations unfactored to determine the action effects imposed through the soil (S_e) for—
 - (i) the soil-supporting structure as a whole for geotechnical strength design, e.g., active pressure on a retaining wall, or earth pressure on a buried structure; and
 - (ii) each component of the structure for structural strength design, e.g., bending moments or shear forces.

NOTE: As an example, for geotechnical strength design of a retaining wall, the action effects would include the earth pressure arising from dead loading, surcharge loading, pressures arising from compaction, earthquake loading and water pressure. For geotechnical strength design of a buried structure, the action effects would include both vertical and lateral earth pressures arising from the above sources.

- (d) The ultimate geotechnical strength (R_{ug}) shall be determined in accordance with Clause 13 or 14, as appropriate, using unfactored characteristic values of material parameters.
- (e) The design geotechnical strength, e.g., passive resistance on a retaining wall, shall be determined using the ultimate geotechnical strength (R_{ug}) multiplied by a geotechnical strength reduction factor (ϕ_g).

The structure shall be proportioned so that—

$$\phi_g R_{ug} \geq S^* \quad \dots 7.3.3(1)$$

where S^* is equal to $1.0S_e$ for geotechnical strength design and ϕ_g is selected in accordance with Clause 7.3.5.

NOTE: ϕ_g for soil-supporting structures takes into account the load factors being 1.0.

- (f) The design structural strength for each structural component shall be determined in accordance with AS 5100.5 or AS 5100.6, as appropriate, by multiplying the ultimate structural strength (R_{us}) by the appropriate strength reduction factor (ϕ_s).

Each of the structural components shall be proportioned so that—

$$\phi_s R_{us} \geq S^* \quad \dots 7.3.3(2)$$

where S^* is equal to $1.5S_e$ for structural strength design, unless required otherwise.

7.3.4 Characteristic values

Characteristic values of the soil and rock parameters shall be selected, based on the following considerations:

- (a) Geological and geotechnical background information.
- (b) The possible modes of failure.
- (c) Results of laboratory and field measurements, taking into account the accuracy of the test method used.

- (d) A careful assessment of the range of values that might be encountered in the field.
- (e) The ranges of in situ and imposed stresses likely to be encountered in the field.
- (f) The potential variability of the parameter values and the sensitivity of the design to these variabilities.
- (g) The extent of the zone of influence governing the soil behaviour, for the limit state being considered.
- (h) The influence of workmanship on artificially placed or improved soils.
- (i) The effects of construction activities on the properties of the in situ soil.

NOTES:

- 1 In general, the characteristic value of geotechnical parameter should be a conservatively assessed value of that parameter. Engineering judgement needs to be exercised in making such an assessment, with geotechnical engineering advice being obtained as required.
- 2 Many soil parameters are not constants, but depend on factors such as the level of stress or strain, the mode of deformation, drainage conditions, moisture contents and their variations over time.
- 3 It should be recognized that a low characteristic value of a geotechnical parameter is not always necessarily a conservative value. For example, in cases involving dynamic or earthquake loads, conservatism may require the selection of a high value of a particular parameter. The sensitivity of the calculated result to the relevant parameter should be taken into consideration.
- 4 Bending moments in buried structures are sensitive to the relative stiffness of the structure and the surrounding soil. The design should consider variation in the stiffness parameters of both the soil and the structure.
- 5 Except where specifically noted, the term soil includes soil and rock. In many cases, weak weathered rock can be analyzed as for soil; however, special techniques may be required for the analysis of strong rock.

7.3.5 Geotechnical strength reduction factors (ϕ_g)

The geotechnical strength reduction factors specified in this Standard shall be used, taking into account the following:

- (a) Methods used to assess the geotechnical strength.
- (b) Variations in the soil conditions.
- (c) Imperfections in construction.
- (d) Nature of the structure and the mode of failure.
- (e) Importance of the structure and consequences of failure.
- (f) Standards of workmanship and supervision of the construction.
- (g) Load variations and cyclic effects.

Values of ϕ_g for specific cases are set out in Clauses 10, 11, 12, 13 and 14.

The geotechnical strength reduction factors selected shall be approved by the relevant authority.

7.4 Design for stability

The structure as a whole, and each of its elements, including the foundations, shall be designed to prevent instability due to overturning, uplift or sliding, as follows:

- (a) Loads determined in accordance with Clause 8.2 shall be subdivided into components tending to cause instability and components tending to resist instability.

- (b) The design action effects (S^*) shall be calculated from the components of the load tending to cause instability, using the load combinations specified in Clause 8.3.
- (c) The ultimate resistance shall be calculated as set out in Clauses 10, 11, 12, 13 and 14. The design resistance shall be computed by multiplying the ultimate resistance by the appropriate strength reduction factor.
- (d) The whole or part of the structure shall be proportioned so that the design action effects are less than or equal to the design resistance.

7.5 Design for serviceability

Foundations and soil-supporting structures shall be designed for serviceability by controlling or limiting settlement, horizontal displacement and cracking.

Under the load combinations for serviceability design specified in Clause 8.4, deflections and horizontal displacements shall be limited to ensure that the foundations and the structure remain serviceable over their design lives.

7.6 Design for strength, stability and serviceability by load testing a prototype

Notwithstanding the requirements of Clauses 7.3, 7.4 and 7.5, foundations or soil-supporting structures may be designed for strength, stability or serviceability by load testing using appropriate test loads. If this alternative procedure is adopted, the requirements for durability (see Clause 9) and other relevant design requirements (see Clause 7.8) shall still apply.

7.7 Design for durability

Foundations and soil-supporting structures shall be designed for durability in accordance with Clause 9.

7.8 Design for other relevant design requirements

Any special design criteria, such as scour, fatigue, flood or collision loading, cyclic loading or liquefaction arising from seismic actions shall be considered. Where relevant, these design criteria shall be taken into account in the design of the foundation or the structure in accordance with the principles of the Standard.

When designing new foundations close to existing structures, the effect of the new structure on existing work, during construction and subsequently, shall be considered. The effect of possible future developments on the proposed work after it is completed shall also be considered, if required by the relevant authority.

NOTE: Some of the circumstances specified in this Clause may lead to either additional loadings (in the case of floods and collisions), a reduction in the depth of soil-resisting loadings (in the case of scour), or to a reduction in soil strength and stiffness (in the case of scour, flood, fatigue, cyclic loading and liquefaction), or a combination of these effects. In the case of collision loading, the rapid rate of load application may provide a basis to adopt increases in the design strength and stiffness of the soil, but such increases are generally ignored for the purposes of design.

8 LOADS AND LOAD COMBINATIONS

8.1 General

The loads and load combinations for strength, stability and serviceability design shall be as specified in Clauses 8.2, 8.3 and 8.4.

8.2 Loads

8.2.1 General

The design for ultimate and serviceability limit states shall take into account the appropriate action effects arising from the following:

- (a) All loads and other actions specified in AS 5100.2.

- (b) Soil movement resulting from slip, reactive soils, consolidation, heaving and other vertical and lateral earth movements.
- (c) Loads from surcharges.
- (d) Distribution of wheel loads through fill.
- (e) Water pressure loads and seepage forces.
- (f) Increase in loads on buried structures because of differential soil movements.
- (g) Compaction pressures.
- (h) Displacement pressures from piling.
- (i) Any additional loads and actions that may be applied.

8.2.2 *Loads induced by soil movement*

Allowance shall be made for loads induced by soil movements as follows:

- (a) Where foundations are situated in soil undergoing settlement, allowance shall be made for loads (F_{nf}) resulting from negative friction on the foundation.
- (b) Where foundations are situated in expansive soils, such as reactive clays or those subject to frost action, allowance shall be made for the compressive and tensile loads (F_{es}) which may develop in the foundation, structure or its elements.
- (c) Where foundations are subject to lateral ground movements, allowance shall be made for bending moments, shear forces and axial loads (F_{em}) induced by such movements.
- (d) Where heave may arise because of unloading of the ground as a result of excavation, allowance shall be made for the bending moments, shears and axial forces (F_{em}) induced by the resulting ground movements.

NOTE: Consideration should be given to each of the following conditions when earth pressure loads on retaining structures are being determined:

- (a) Configuration, nature and drainage properties of the backfill material.
- (b) Displacement characteristics of the wall.
- (c) Interface conditions between the wall and the backfill.
- (d) Method of compaction of the backfill material.
- (e) Sequences of excavation and placement of anchorages and struts.

In spill-through abutments, to take into account the possible arching of fill between columns, one of the following procedures shall be adopted:

- (i) A detailed ground-structure interaction analysis shall be carried out to determine the earth pressures acting on the columns.
- (ii) In the absence of a detailed analysis, no reduction of earth pressure loading shall be made, to allow for a space between columns if that space is less than twice the width across the back of the columns. For greater spacings, friction on the sides of the columns or counterforts shall be considered and the earth pressure loading on each column shall be taken on an equivalent width not less than twice the actual width across the back of the columns.

8.2.3 *Construction loads*

Loads and actions that arise from construction activities shall be evaluated, and those that affect the requirements for strength, stability or serviceability shall be taken into account.

8.2.4 *Water pressure*

The loads applied by hydrostatic pressure of water or ground water seepage forces, or both, shall be taken into account in the design of foundations and soil-supporting structures. The effects of buoyancy on the structural components and on soil shall be included.

8.3 **Load combinations for strength and stability design**

8.3.1 *General*

The load combinations for strength and stability design shall be as specified in Clauses 8.3.2 and 8.3.3.

8.3.2 *Foundations*

For foundations where the loads are imposed predominantly from the structure or from loads applied to the structure, the load combinations shall be as follows:

- (a) The design loads for a foundation shall be the combination of factored loads that produces the most adverse effect on the foundation in accordance with AS 5100.2.
- (b) If there are loads caused by soil movements (see Clause 8.2.1), the loads shall be considered as permanent effects and shall be factored and combined with the other load combinations specified in AS 5100.2 as follows:
 - (i) For structural strength and stability design, the loads caused by soil movements shall be factored as follows:
 - (A) $1.2F_{nf}$ — For negative friction loads caused by consolidation of the surrounding soil.
 - (B) $1.5F_{es}$ — For compressive and tensile loads caused by vertical soil movements other than consolidation of the surrounding soil.
 - (C) $1.5F_{em}$ — For bending moments, shear forces and axial loads caused by lateral soil movements and heave.
 - (ii) For geotechnical strength design, the possibility of soil movements altering the ultimate geotechnical strength shall be considered.

NOTE: Usually, soil movements have little or no effects on ultimate geotechnical strength of foundations; however, soils susceptible to strain-softening may be affected.

Where other additional loads and actions are to be applied and no load factor is given in AS 5100.2 for these loads and actions, a load factor not less than 1.5 shall be adopted for both structural and geotechnical design.

8.3.3 *Soil-supporting structures*

For soil-supporting structures where the loads are imposed predominantly from the soil, the design loads and other actions for strength and stability design of a soil-supporting structure shall be the combination of loads that produces the most adverse effect on the structure in accordance with AS 5100.2. The loads shall be combined using a load factor of 1.0 for each of the loads.

8.4 **Load combinations for serviceability design**

The design loads and other actions for serviceability design of foundations and soil-supporting structures shall be taken from the appropriate combination of factored loads in accordance with AS 5100.2. The design loads shall include loads resulting from soil movements and other additional loads specified in Clause 8.2, where appropriate, using a load factor of 1.0 for each of these loads.

9 DURABILITY

9.1 General

The objective of the design of the structure with respect to durability shall be—

- (a) to achieve, with appropriate maintenance, the specified service life; and
- (b) that all the specified design criteria continue to be satisfied throughout the service life.

Consideration shall be given to the possibility of deterioration of structural components of foundations and soil-supporting structures as a result of aggressive substances in soils or rocks, in ground water, seawater and water in streams. Account shall also be taken of the abrading effects of waterborne sands and gravels.

In addition, other specific durability criteria may apply, as required by the relevant authority.

9.2 Durability of timber

Untreated timber shall not be used as permanent components of foundations or soil-supporting structures unless permitted by the relevant authority. Any untreated timber shall be located below the permanent ground water level. Where borers exist, untreated timber shall not be used in marine conditions.

Where permitted by the relevant authority, suitably treated timber of durable species may be used as permanent components of foundations or soil-supporting structures, but its use shall be limited, having due regard to consequences of failure and replacement and the degree to which the treatment is effective over the entire cross-section.

NOTE: The use of timber in foundations and soil-supporting structures should be limited to temporary structures or to the repair of existing timber structures.

9.3 Durability of concrete

The requirements for design for durability of concrete components of foundations and soil-supporting structures given in AS 5100.5 shall apply.

For buried concrete structures where stray currents are likely to be present, e.g., adjacent to electrified railway lines, action shall be taken as required by the relevant authority to prevent corrosion of the reinforcement.

9.4 Durability of steel

Unless more site-specific information is available and unless required otherwise by the relevant authority, the following rates of corrosion for unprotected steel surfaces shall be used for design purposes:

- (a) 1.5 mm total for the life of the structure for each face in contact with soil, above and below ground water, provided the soil is undisturbed or comprises compacted, well-graded, chemically neutral, structural fill.
- (b) 0.025 mm per year for each face in contact with open-graded or rubble fill, or sands and gravels that have moving ground water.
- (c) 0.05 mm per year for each face exposed to fresh water and not in contact with soil.
- (d) 0.08 mm per year for each face exposed to seawater, except in the splash zone where twice this rate shall be used.

NOTES:

- 1 The presence of high concentrations of chloride ions, oxygen and sulphate-reducing bacteria are significant in determining the level of corrosion to steel surfaces.
- 2 Buried or immersed steel surfaces may be protected by galvanizing or coating with various materials including bitumen, flake-filled polyesters, epoxy mastics, polyethylene and others. The expected life of the galvanizing or coatings should be taken into account in the design.

For steel surfaces exposed to the atmosphere, the rate of corrosion will depend on the type of protective coating, level of stress, structural details, the extent of routine maintenance and atmospheric conditions. The rate of corrosion to be adopted shall be as required by the relevant authority.

For buried steel structures where stray currents are likely to be present, e.g., adjacent to electrified railway lines, action as required by the relevant authority shall be taken to minimize corrosion.

9.5 Durability of slip layers

Slip layer coatings applied to piling shall be as approved by the relevant authority.

9.6 Durability of other materials

Where foundations or soil-supporting structures are to be constructed from materials other than those covered specifically by this Standard, reference shall be made to other appropriate Standards and current technical literature for material-specific information on durability. Where possible, durability of such materials shall be assessed using testing appropriate to the particular situation.

The durability of other materials shall be as required by the relevant authority.

10 SHALLOW FOOTINGS

10.1 Scope

This Clause applies to all types of shallow footings, such as pad, strip and raft footings for structures and retaining walls. For the purpose of this Standard, a shallow footing is one that is founded at shallow depth and where the contribution of the strength of the ground above the footing level does not influence the bearing resistance significantly.

10.2 Load and load combinations

Shallow footings shall be designed for the loads and other actions set out in Clause 8.2.

The load combinations for strength, stability and serviceability shall be as specified in Clauses 8.3.2 and 8.4.

10.3 Design requirements

10.3.1 General

The magnitude and disposition of the structural loads and actions, and the bearing resistance of the ground, shall be considered when selecting the appropriate type of shallow footing.

The footing shall be designed to satisfy the strength design requirements set out in Clause 7.3.2 and the serviceability design requirements set out in Clause 7.5.

10.3.2 Footing depth and size

When determining the footing depth, the following shall be considered:

- (a) The depth of an adequate bearing stratum.
- (b) The effects of scour.
- (c) In the case of clay soils, the depth of appreciable ground movement caused by shrinkage and swelling due to moisture changes resulting from seasonal variations or trees and shrubs.
- (d) The depth to which frost heave is likely to cause appreciable ground movements.
- (e) Subsequent nearby construction work such as trenches for services.
- (f) Possible ground movements.

- (g) The level of the ground water table and the problems that may occur if excavation for the foundation is required below this level.

When determining the footing width, consideration should be given to issues related to practical excavation constraints, setting-out tolerances, working space requirements and the dimensions of the substructure supported by the footing.

10.3.3 *Design for geotechnical strength*

10.3.3.1 *General*

Ultimate limit states corresponding to a mechanism in the ground or rupture of a critical section of the structure because of ground movements shall be evaluated using the ultimate limit state actions and loads, and the ultimate resistance factored by an appropriate strength reduction factor.

10.3.3.2 *Overall stability*

Consideration shall be given to the possibility of failure resulting from loss of overall stability.

The design resistance for stability failure of the ground mass shall be not less than the design strength effect of any possible modes of failure.

NOTE: Situations in which overall stability may be particularly important include—

- (a) footings near or on an inclined site, a natural slope or an embankment;
- (b) footings near an excavation or a retaining structure;
- (c) footings near a river, canal, lake, reservoir or the sea shore; and
- (d) footings near mine workings or buried structures.

10.3.3.3 *Ultimate bearing failure*

Footings subjected to vertical or inclined loads or overturning moments shall be proportioned such that the design bearing capacity is greater than or equal to the design action effect (S^*), i.e.—

$$\phi_g R_{ug} \geq S^* \quad \dots 10.3.3.3$$

where

ϕ_g = geotechnical strength reduction factor

R_{ug} = ultimate geotechnical strength (bearing capacity) of the footing

In assessing S^* , allowance shall be made for the weight of the footing and any backfill material on the footing.

The value of R_{ug} shall be established by using the results of field or laboratory testing of the ground. Allowance shall be made for the effects of the following:

- (a) Variations in the level of the ground water table and rapid draw down.
- (b) Any weak or soft zones in the soil or rock below the founding level.
- (c) Unfavourable bedding or jointing of rock strata, especially in sloping ground.
- (d) Possible influence of time effects and transient, repeated or vibratory loads on the soil shear strength.
- (e) Load eccentricity and inclination. In assessing the ultimate geotechnical strength (R_{ug}) of footings subjected to eccentric loads, allowance shall be made for the possibility of very high edge stresses and a reduced effective contact area between the footing and the ground as a result of load eccentricity.
- (f) Presence of sloping ground or nearby excavations.

NOTE: The ultimate bearing capacity of a footing may be estimated analytically by using soil shear strengths measured in appropriate laboratory or field tests, or by using empirical or quasi-analytical relationships developed from the results of in situ tests such as the standard penetration test, the static cone penetration test, the plate loading test, the vane shear test or the pressuremeter test.

The geotechnical strength reduction factor (ϕ_g) shall be selected in accordance with Clause 7.3.5, and Tables 10.3.3(A) and 10.3.3(B).

TABLE 10.3.3(A)
RANGE OF VALUES OF GEOTECHNICAL STRENGTH REDUCTION FACTOR (ϕ_g) FOR SHALLOW FOOTINGS

Method of assessment of ultimate geotechnical strength	Range of values of ϕ_g
Analysis using geotechnical parameters based on appropriate advanced in situ tests	0.50–0.65
Analysis using geotechnical parameters from appropriate advanced laboratory tests	0.45–0.60
Analysis using CPT tests	0.40–0.50
Analysis using SPT tests	0.35–0.40

NOTE: Examples of testing regimes are given in AS 5100.3 Supp 1.

TABLE 10.3.3(B)
GUIDE FOR ASSESSMENT OF GEOTECHNICAL STRENGTH REDUCTION FACTOR (ϕ_g) FOR SHALLOW FOOTINGS

Lower end of range	Upper end of range
Limited site investigation	Comprehensive site investigation
Simple methods of calculation	More sophisticated design method
Limited construction control	Rigorous construction control
Severe consequences of failure	Less severe consequences of failure
Significant cyclic loading	Mainly static loading
Foundations for permanent structures	Foundations for temporary structures
Use of published correlations for design parameters	Use of site-specific correlations for design parameters

10.3.3.4 Failure by sliding

Footings subjected to horizontal loads shall be proportioned such that the design action effect (S^*) shall satisfy the following:

$$\phi_g H_{ug} + \phi_g E_{pr} \geq S^* \quad \dots 10.3.3.4$$

where

H_{ug} = ultimate shear resistance at the base of the footing

E_{pr} = ultimate passive resistance of the ground in front of the footing

ϕ_g = geotechnical strength reduction factor, which shall be selected in accordance with Clause 7.3.5, and Tables 10.3.3(A) and 10.3.3(B)

NOTE: The values of both $H_{ug}\phi_g$ and $E_{pr}\phi_g$ should be related to the scale of movement anticipated under the limit state being considered. For large movements associated with ultimate limit states, the possible relevance of post-peak softening behaviour should be considered.

For foundations on clay soils bearing within the zone of seasonal movements, the possibility that the clay could shrink away from the vertical faces of foundations shall be considered.

The possibility that the soil in front of the foundation may be removed by erosion or human activity shall be considered.

10.3.4 Design for structural strength

10.3.4.1 General

The design structural strength (R_s^*) of the footing shall satisfy the following:

$$R_s^* \geq S^* \quad \dots 10.3.4.1$$

where

$$R_s^* = \phi_s R_{us}$$

ϕ_s = structural strength reduction factor

R_{us} = ultimate structural strength

ϕ_s shall be obtained from AS 5100.5 or AS 5100.6, as appropriate.

When calculating R_{us} for strip footings or raft footings, consideration shall be given to the distribution of soil pressure at the base of the footing.

10.3.4.2 Structural failure as a result of footing movement

Differential vertical and horizontal displacements of a footing or between footings under the serviceability limit state design actions and ground deformation parameters shall be considered. The footing shall be designed such that these displacements do not lead to an ultimate limit state occurring in the supported structure.

10.3.5 Design for serviceability limit states

10.3.5.1 General

Consideration shall be given, as appropriate, to the following:

- (a) The displacement of a single footing.
- (b) Displacements and differential displacements of footing groups, footing beams or rafts.
- (c) Vibrations arising from repetitive, vibratory or dynamic loads.

Footing displacements shall be calculated using the serviceability loads and actions. Calculated footing displacements shall satisfy the following:

- (i) The displacement shall not be greater than the serviceability limit displacement.
- (ii) The differential displacement shall not be greater than the serviceability limit value.

The serviceability limit values of displacement and differential displacement shall be selected such that they do not result in detrimental effects on the structure being supported.

In estimating the displacements, consideration shall be given to the following components of displacement:

- (A) Immediate displacement.
- (B) Time-dependent displacements caused by soil consolidation.
- (C) Long-term soil creep displacements.

Any possible additional settlement caused by self-compaction of the soil shall also be assessed.

The differential settlements and relative rotations shall be assessed, taking account of both the distribution of loads and the possible variability of the ground.

NOTE: Differential settlements calculated without taking account of the stiffness of the structure tend to be overpredictions. An analysis of ground-structure interaction may be used to justify reduced computed values of differential settlements.

Characteristic values of soil deformation design parameters for use in analysis of footing displacements for the serviceability limit state shall be assessed on the basis of appropriate laboratory tests or field tests, or by evaluating the behaviour of neighbouring similar structures. A geotechnical reduction factor need not be applied to the parameters so assessed.

NOTES:

- 1 In general, the characteristic value of a geotechnical parameter should be a conservatively assessed value of that parameter. Engineering judgement needs to be exercised in making such an assessment.
- 2 Footing displacements can be estimated from various methods, including—
 - (a) analysis using elastic theory, using appropriate parameters for immediate and long-term displacements;
 - (b) analysis using consolidation theory, which is useful for clay soils where there is a relatively large time-dependent displacement component due to consolidation;
 - (c) analysis using appropriate soil constitutive models, usually via finite element analysis; and
 - (d) analysis using results from in situ tests, which may include both analytical techniques and empirical methods (applied mainly to sandy soils).

10.3.5.2 Tilting

The calculated tilt of the footing shall not be greater than the serviceability limit value for proper functioning of the supported structure.

In the case of footings subject to loads with large eccentricities, measures shall be adopted to avoid ‘doming’ of the ground surface beneath the footing, which may cause rocking of the footing.

NOTE: Situations that may cause significant tilting include—

- (a) eccentric loads;
- (b) inclined loads;
- (c) non-uniform soil conditions; and
- (d) overturning moments.

10.3.6 Design for durability

Durability requirements shall be considered as set out in Clause 9. Where materials other than concrete and steel are to be used for the construction of the shallow footing, the requirements for durability in the relevant Standard for that material shall apply, unless otherwise specified by the relevant authority.

Where no Standard applies to the materials used in the shallow foundation, the requirements of the relevant authority shall apply.

10.4 Structural design and detailing

Structural design and detailing for shallow footings built of concrete and steel shall be in accordance with AS 5100.5 or AS 5100.6, as appropriate.

Where materials other than concrete and steel are to be used for the construction of the structure, then the requirements of the relevant Standard for that material shall apply to the structural design and detailing of the structure, unless otherwise specified by the relevant authority.

Where no Standard applies to the materials used for the construction of the structure, the requirements of the relevant authority shall apply.

10.5 Materials and construction requirements

Materials and construction requirements for shallow foundations built of concrete and steel shall be in accordance with AS 5100.5 or AS 5100.6, as appropriate.

Where materials other than concrete and steel are to be used for the construction of the structure, then the requirements of the relevant Standard for that material shall apply, unless otherwise specified by the relevant authority.

Where no Standard applies to the materials used for the construction of the structure, the requirements of the relevant authority shall apply.

11 PILED FOUNDATIONS

11.1 Scope

This Clause sets out minimum requirements for the design, construction and testing of piled foundations. The provisions apply to axially and transversely loaded displacement and non-displacement piles installed by driving, jacking, screwing or boring with or without grouting.

11.2 Load and load combinations

Loads and load combinations for pile design shall be in accordance with AS 2159 except where specified otherwise in Clause 8.

11.3 Design requirements

11.3.1 General

Pile design requirements and procedures shall be in accordance with AS 2159 except where specified otherwise in Clause 7.

The geotechnical design of piles and geotechnical strength reduction factors shall be in accordance with AS 2159. The range of geotechnical strength reduction factors for piles shall be as given in Appendix A.

11.3.2 Design for strength

Structural design for steel and concrete piles shall be in accordance with AS 2159 except where specified otherwise in AS 5100.5 and AS 5100.6.

Where the use of timber piles is permitted by the relevant authority, timber piles shall be designed in accordance with AS 2159.

11.3.3 Design for serviceability

For the serviceability design of piled foundations, the provisions of Clause 7.5 shall apply. In estimating the settlement and horizontal displacements, account shall be taken of the stiffness of the ground and structural elements, and of the sequence of construction.

The permissible displacements for the piled foundations shall be established, taking into account the tolerance to deformation of the supported structure and services.

11.3.4 Design for durability

Design for durability shall be in accordance with AS 2159 except where specified otherwise in AS 5100.5 and AS 5100.6. Where materials other than concrete and steel are to be used, the requirements for durability in the relevant Standard for that material shall apply, unless otherwise specified by the relevant authority.

Where no Standard applies to the materials used, the requirements of the relevant authority shall apply.

11.4 Structural design and detailing

11.4.1 General

Structural design and detailing for steel and concrete piles shall be in accordance with AS 2159, except where specified otherwise in AS 5100.5 or AS 5100.6, as appropriate.

Where materials other than concrete and steel are to be used for construction of the pile, then the requirements of the relevant Standard for that material shall apply to the structural design and detailing of the pile, unless otherwise specified by the relevant authority.

Where no Standard applies to the materials used for the construction of the pile, the requirements of the relevant authority shall apply.

Piles shall be designed as structural columns with the degree of end fixity and lateral support appropriate to the surrounding soil conditions and the behaviour of the structure. The effects of scour in removing lateral support shall be considered. Unless approved otherwise by the relevant authority, piles subjected to lateral loads or bending moment shall be designed to provide a design resistance greater than or equal to the maximum serviceability and ultimate design action effects for a distance at least 2 m below the point where lateral support commences.

The use of pile splices shall be limited to situations where their use is unavoidable.

In addition to considerations relevant to the design of piles as structural members, the design of specific types of piles shall take into account the requirements set out in Clause 11.4.2, as appropriate.

11.4.2 Design details relevant to specific types of piles

11.4.2.1 Precast reinforced concrete piles

For precast reinforced concrete piles, the following shall apply:

- (a) *Size and shape* The cross-sectional area shall be not less than 90 000 mm² except that where the pile is in salt water, it shall be not less than 140 000 mm². Any square corners shall have a 25 mm chamfer. The ends of the pile shall be at right angles to the pile axis. Any taper shall be concentric with the axis of the pile, with a pile diameter or average least width measured 600 mm above the toe of not less than 200 mm.
- (b) *Driving straps* The head of a reinforced concrete pile shall be reinforced with a steel strap a minimum of 6 mm thick and 75 mm wide cast with the pile to minimize spalling under hard driving conditions.
- (c) *Reinforcement* Longitudinal reinforcement, consisting of not less than four bars spaced uniformly around the perimeter of the pile, shall be provided in all cases, except that if more than four bars are used, the number may be reduced to four in the bottom 1.2 m of the pile.

The area of longitudinal reinforcement shall be not less than 1.0% of the cross-sectional area of the pile.

Joints in longitudinal reinforcement shall be avoided if possible. Where required, such joints shall be made by butt welding in accordance with AS/NZS 1554.3.

The full length of the longitudinal reinforcement shall be enclosed with stirrups or helical reinforcement of not less than 6 mm diameter. The volume of the stirrups or helical reinforcement shall not be less than 0.2% of the gross volume of the pile, with a spacing or pitch of not more than half the average least width, or diameter, of the pile.

For a distance from each end of the pile of not less than two times the average least pile width, or diameter, the volume of the stirrups or helical reinforcement shall be not less than 0.4% of the volume of that part of the pile. For the spacing of the stirrups or helical reinforcement, the transition from the close spacing at the ends of the pile to the larger spacing shall not be less than two times the average least pile width.

- (d) *Mechanical joints* Mechanical joints shall only be used with the approval of the relevant authority. Precast pile lengths mechanically joined shall be not less than 3 m and not more than 20 m long. The mechanical joints shall be designed so that they provide a permanent connection between the pile lengths. The strength of the joint shall be not less than that of the lengths of pile being joined. Durability of mechanical joints shall comply with Clause 9.

11.4.2.2 *Prestressed concrete piles*

For prestressed concrete piles, the following shall apply:

- (a) *Concrete strength* The concrete shall have a 28 day compressive strength of not less than 40 MPa.
- (b) *Size and shape* The provisions of Clause 11.4.2.1(a) shall apply. Piles with a diameter or average least width less than 450 mm shall be solid. Larger diameters or average least widths may be hollow.
- (c) *Prestress and reinforcement* Prestressing tendons shall be provided, spaced uniformly around the perimeter of the pile. The minimum residual compressive stress shall be 7 MPa.

Non-prestressed longitudinal reinforcement shall be provided as required for driving, splicing and anchorage to pile caps.

Helical reinforcement or stirrups shall be provided as set out in Clause 11.4.2.1(c) except that for hollow piles the volume of helical reinforcement or stirrups in the body of the pile shall not be less than 0.3% of the gross pile volume and for solid piles, not less than 0.2%.

- (d) *Mechanical joints* The provisions of Clause 11.4.2.1(d) shall apply.

11.4.2.3 *Cast-in-place reinforced concrete piles*

For cast-in-place reinforced concrete piles, the following shall apply:

- (a) *Reinforcement* Stirrups or helical reinforcement shall have a spacing or pitch not greater than 150 mm.
Longitudinal reinforcement shall be placed equally spaced around the perimeter of the pile and shall extend the full depth of the pile. The clear spacing between longitudinal bars shall not be less than 75 mm, including bars at lapped splices.
- (b) *Casing* Steel casings provided for ground support or inspection purposes shall have a minimum thickness of 10 mm. Welding of casings shall be in accordance with AS/NZS 1554.1.

11.4.2.4 *Steel piles*

Steel piles shall have a minimum thickness of 10 mm at the end of the design life after taking into account corrosion. Welding of steel piles shall be in accordance with AS/NZS 1554.1.

11.5 Materials and construction requirements

11.5.1 *General*

Piles shall be constructed in accordance with AS 2159, except where specified otherwise in AS 5100.5 or AS 5100.6, as appropriate.

Where materials other than concrete and steel are to be used for construction of the pile, then the requirements of the relevant Standard for that material shall apply, unless otherwise specified by the relevant authority.

Where no Standard applies to the materials used for the construction of the pile, the requirements of the relevant authority shall apply.

11.5.2 *Spacing, edge distance and embedment of piles*

For friction piles, the spacing centre-to-centre shall not be less than 2.5 times the diameter or nominal size of the pile. For piles deriving their resistance mainly from end-bearing, the spacing centre-to-centre shall be not less than twice the size of the pile. For piles with rakes or enlarged bases, increased spacings may be required to suit the geometry and clearances.

The distance from the outside of any pile in a pile group to the edge of a concrete pile cap shall be a minimum of 100 mm after taking into account construction tolerances.

The embedment of the concrete of a concrete pile into a concrete pile cap shall be a minimum of 50 mm.

11.6 Testing

Static, dynamic and integrity testing of piles shall be in accordance with AS 2159, unless required otherwise by the relevant authority.

12 ANCHORAGES

12.1 Scope

The provisions in this Clause apply to any type of anchorage used to restrain a structure by transmitting a tensile force to a load bearing formation of soil or rock.

NOTES:

- 1 Anchorages may include—
 - (a) non-prestressed ties and anchors, e.g., deadman anchors, soil nails, sheet piles, raked piles; and
 - (b) prestressed anchorages, e.g., post-tensioned soil and rock ground anchors.
- 2 Anchorages may be employed as temporary or permanent elements of a structure.

12.2 Load and load combinations

Anchorages shall be designed for the loads and other actions set out in Clause 8.2.

The load combinations for strength, stability and serviceability shall be as specified in Clauses 8.3 and 8.4.

12.3 Design requirements

12.3.1 *General*

Anchorage design shall take into account all foreseeable circumstances during the design life of the anchorage. The corrosion and creep of the permanent anchorages shall be considered.

NOTE: Preferably, anchorage systems for which successful long-term experience has been documented with respect to performance and durability should be used.

12.3.2 Site investigation

Site investigations prior to the design and construction of anchorages shall include ground formations outside the construction site if effects of the anchorage forces will occur there.

12.3.3 Design for strength

For the geotechnical and structural strength design of anchorages, the provisions of Clause 7.3 shall apply.

For ground anchors, the design geotechnical strength and the design structural strength shall be calculated as the appropriate ultimate strength (R_u) multiplied by an importance category reduction factor (ϕ_n) given in Table 12.3.3(A) and the appropriate strength reduction factor (ϕ_s) or (ϕ_g). The structural strength reduction factor (ϕ_s) shall be obtained from AS 5100.5 or AS 5100.6, as appropriate. The geotechnical strength reduction factor (ϕ_g) shall be selected in accordance with Clause 7.3.4, and Tables 12.3.3(B) and 12.3.3(C).

To check anchorage strength limit states, three failure mechanisms shall be analysed as follows:

- (a) The failure of the tendon or anchor head in terms of the material strength or failure of bonding at internal interfaces.
- (b) The failure of the anchorage at the tendon-grout or grout-ground interface.
- (c) The overall stability failure of the structure, including the anchorages.

TABLE 12.3.3(A)
IMPORTANCE CATEGORY REDUCTION FACTOR (ϕ_n)

Anchor category	Importance category reduction factor (ϕ_n)
1 Temporary anchors where the service life is less than six months and failure would have few serious consequences and would not endanger public safety, for example, short-term pile test loading using anchors as a reaction system	1.0
2 Temporary anchors with a service life of up to two years where, although the consequences of local failure are quite serious, there is no danger to public safety without adequate warning, for example, retaining wall tie backs	0.93
3 Any permanent anchors and also temporary anchors where the consequences of failure are serious, for example, temporary anchors for main cables of a suspension bridge, or as a reaction for lifting structural members	0.7

TABLE 12.3.3(B)
RANGE OF VALUES OF GEOTECHNICAL STRENGTH
REDUCTION FACTOR (ϕ_g) FOR ANCHORAGES

Method of assessment of ultimate geotechnical strength	Range of values of ϕ_g	
	Permanent structures	Temporary structures
Analysis using the results of site-specific anchorage pull-out tests	0.55	0.60–0.70
Analysis using the results of anchorage pull-out tests in similar ground conditions	0.55	0.55–0.65
Analysis using geotechnical parameters based on appropriate advanced in situ tests	0.5–0.55	0.50–0.65
Analysis using geotechnical parameters from appropriate advanced laboratory tests	0.45–0.55	0.45–0.60
Analysis using CPT tests	0.40–0.50	0.40–0.50
Analysis using SPT tests	0.35–0.40	0.35–0.40

NOTE: Examples of testing regimes are given in AS 5100.3 Supp 1.

TABLE 12.3.3(C)
GUIDE FOR ASSESSMENT OF GEOTECHNICAL STRENGTH
REDUCTION FACTOR (ϕ_g) FOR ANCHORAGES

Lower end of range	Upper end of range
Limited site investigation	Comprehensive site investigation
Simple methods of calculation	More sophisticated design method
Limited construction control	Rigorous construction control
Severe consequences of failure	Less severe consequences of failure
Significant cyclic loading	Mainly static loading
Use of published correlations for design parameters	Use of site-specific correlations for design parameters

12.3.4 *Design for other relevant factors*

During the design of anchorages, the following additional factors shall be considered:

- Creep movement of soil.
- Level of ground water table and possibility of changes in that level.
- Provision for drainage.
- Depths of anchorages relative to global stability of the structure.
- Rigidity of the structure being supported.
- Possibility of movement of the structure.
- Group effects.
- In the case of anchorages in soil, the behaviour of the soil due to the anchor loads.
- In the case of anchorages in rock, anisotropy, inhomogeneity, fracturing and discontinuities of the rock.
- Method of installation.
- Geometry of the anchorage.

- (l) Uplift resistance of soil or rock mass.
- (m) Tendon design.
- (n) Rock to grout bond and grout to tendon bond (grouted anchors).
- (o) Soil to tendon bond for soil nails or soil reinforcement.
- (p) Strength of mechanical anchorage for mechanically secured anchors.
- (q) Non-uniform stress distribution, particularly the potential for ‘unzipping’ failure for anchors with long bond lengths.

12.3.5 *Design for serviceability*

For the serviceability design of anchorages, the provisions of Clause 7.5 shall apply. Consideration shall be given, as appropriate, to the following:

- (a) Loss of anchor force by excessive displacement of the anchor head.
- (b) Loss of anchor force as a result of creep and relaxation.
- (c) Failure or excessive deformation of the structure due to anchor forces.

12.3.6 *Design for durability*

Anchorages and all anchorage components shall be designed to meet the requirements of Clause 9. The design life shall be in accordance with AS 5100.1.

Protection against corrosion shall be provided for all permanent anchorages as required to comply with Clause 9.

Where materials other than concrete and steel are to be used for the construction of the anchorage, the requirements for durability in the relevant Standard for that material shall apply, unless otherwise specified by the relevant authority.

Where no Standard applies to the materials used for the anchorage, the requirements of the relevant authority shall apply.

12.4 Materials and construction requirements

Materials and construction requirements for anchorages shall be in accordance with AS 5100.5 or AS 5100.6, as appropriate.

Where materials other than concrete and steel are to be used for the anchorage, the requirements of the relevant Standard for that material shall apply, unless otherwise specified by the relevant authority.

Where no Standard applies to the materials used in the anchorage, the requirements of the relevant authority shall apply.

12.5 Anchorage installation plan

An anchorage installation plan shall be prepared as part of the technical construction specification for the anchorage system to be used.

An anchorage installation plan shall contain the following information, as appropriate:

- (a) Anchorage type.
- (b) Number of anchorages.
- (c) Location and orientation of each anchorage and tolerances in position.
- (d) Anchorage length.
- (e) Installation sequence for the anchorages.

- (f) For grouted anchorages, the grout material specification, pressure, grouted volume, grouted length and grouting time.
- (g) Required serviceability load for each anchorage.
- (h) Method of corrosion protection.
- (i) Installation technique, such as drilling, placing, bonding and stressing.
- (j) Any other constraints on anchoring activities.

12.6 Anchorage testing

12.6.1 General

The method used for the installation of anchorages subjected to on-site assessment tests shall be fully documented and shall meet the requirements of the relevant authority.

NOTE: A typical generic procedure for the on-site assessment tests of anchorages is described in Appendix B.

Between the time of installation of an anchor and the beginning of a load test, adequate time shall be allowed to ensure that the required quality of the bond at the tendon-grout interface or, where relevant, grout-encapsulation, and grout-ground interface, is achieved.

All measuring apparatus used for anchor testing shall be appropriately sensitive and accurate, and shall be calibrated prior to the testing.

The load-carrying capacity of a grouted anchor shall be evaluated from test results.

The following load tests on anchors shall be carried out on site:

- (a) Proof load tests in accordance with Clause 12.6.2.
- (b) Acceptance tests in accordance with Clause 12.6.3.

12.6.2 Proof load tests

Proof load tests shall be carried out in advance of the construction or on selected working anchors during construction, to assess the capability of the anchor system to achieve the required resistances under the site ground conditions.

NOTE: The proof load tests also provide criteria for the acceptance tests.

Unless required otherwise by the relevant authority, at least one proof load test shall be carried out for each distinct ground condition and anchor type. Relevant load testing carried out previously may also be taken into account.

On large anchorage projects, unless required otherwise by the relevant authority, the number of proof load tests per each distinct ground condition shall be at least 1% of the total number to be installed for temporary anchors where failure is likely to have relatively minor consequences, and at least 2% of the total number to be installed in the case of permanent anchors or of temporary anchors where there are likely to be severe consequences of failure.

The test duration shall be sufficient to ensure that prestress or creep fluctuations stabilize within tolerable limits.

When deriving the characteristic anchorage resistance (R_{ak}) from the measured anchorage capacity (R_{am}) in one or more proof load tests, an allowance shall be made for the variability of the ground and the variability of the effect of anchorage installation. The systematic and random components of variations in the ground conditions shall be distinguished in the interpretation of the proof load tests. As a minimum, both conditions (a) and (b) given in Table 12.6.2 shall be satisfied using the following equation:

$$R_{ak} = \phi_c R_{am} \quad \dots 12.6.2(1)$$

The measured anchorage capacity (R_{am}) obtained from the proof load tests shall be equal to the lowest of the calculated loads corresponding to the first two failure mechanisms referred in Clause 12.3.3 and the creep limit load.

The anchorage resistance (R_a) shall be derived from the following equation:

$$R_a = \phi_n R_{ak} \quad \dots 12.6.2(2)$$

where ϕ_n shall be as given in Table 12.3.3(A).

The anchorage resistance (R_a) shall satisfy the following condition:

$$R_a \geq S^* \quad \dots 12.6.2(3)$$

where S^* is the design action effect for the anchorage.

The proof load test procedure shall be such that conclusions can be drawn about the anchor capacity, the creep limit load and the apparent free tendon length. Attention shall be paid to the number of loading steps, the duration of these steps and application of the load cycles.

NOTE: Ground variability can be taken into account by considering the different zones of homogeneous conditions or a trend of ground conditions with position on the site. The data about the installation of the anchorages should then be checked, and any deviations from normal installation should be accounted for. Such variations should be covered in part by a correct selection of the anchor for the proof load tests.

TABLE 12.6.2
CONVERSION FACTORS (ϕ_c)

Number of proof load tests	1	2	>2
ϕ_c on mean R_{am}	0.67	0.74	0.77
ϕ_c on lowest R_{am}	0.67	0.8	0.91

12.6.3 Acceptance tests

Acceptance tests shall be carried out to demonstrate that each of the anchorages installed has the capacity to carry the calculated design load.

All grouted anchorages shall be subjected to an acceptance test before they become operational, and prior to lock-off.

Acceptance tests shall be performed using procedures and acceptance criteria derived from the results of the proof load tests with the aim of proving the ability of each anchorage to support the relevant limit state loads as approved by the relevant authority.

The test procedure shall provide confirmation of the apparent free tendon length and confirmation that the tendon relaxation after lock-off will be acceptable. As a minimum, each anchorage shall be loaded to x times the design serviceability load of the anchor, where x is equal to 1.5 for permanent anchorages, and x is equal to 1.3 for temporary anchorages, or as required otherwise by the relevant authority.

NOTE: The acceptance test may be used to pre-load the anchorage in order to minimize future tendon relaxation.

12.7 Monitoring

Where verification of the long-term capacity of the anchorage is required, provision for monitoring or subsequent load testing of the anchorage, or both, shall be provided as part of the design.

13 RETAINING WALLS AND ABUTMENTS

13.1 Scope

The requirements for the design of retaining walls and abutments shall be as set out herein, unless required otherwise by the relevant authority.

NOTE: The design of reinforced soil walls and structures is not covered by this Standard.

13.2 Loads and load combinations

Retaining walls and abutments shall be designed for loads and other actions set out in Clause 8.2.

The load combinations for strength, stability and serviceability shall be as specified in Clauses 8.3.3 and 8.4.

13.3 Design requirements

13.3.1 *Design for strength and stability*

For the geotechnical and structural design of retaining walls and abutments, the provisions of Clause 7.3.3 shall apply. In designing for stability, the provisions of Clause 7.4 shall apply.

As a minimum, the following limit states shall be considered:

- (a) Sliding within or at the base of the structure.
- (b) Rotation of the structure.
- (c) Rupture of a structural element such as a wall, anchor, wale or strut, or failure of the connection between such elements.
- (d) Global failure.
- (e) Bearing failure.

The design geotechnical strength and design structural strength shall be calculated as the appropriate ultimate strength (R_u) multiplied by the appropriate strength reduction factor (ϕ).

The geotechnical strength reduction factor (ϕ_g) shall be selected in accordance with Clause 7.3.5, and Tables 13.3.1(A) and 13.3.1(B), taking into account the limit state considered and shall be subject to the approval of the relevant authority.

The structural strength reduction factor (ϕ_s) shall be obtained from AS 5100.5 or AS 5100.6, as appropriate.

For retaining walls and abutments subjected to differential water pressures, the possibility of failure by hydraulic instability (erosion or piping) shall be considered.

For soil profiles containing fine-grained soils, both short-term and long-term conditions shall be considered.

Where the stability of a retaining wall or abutment depends on the passive resistance of the ground in front of the structure or abutment, the ground level in front of the wall or abutment shall be lowered by an amount Δh . For a cantilever structure, Δh shall be taken as 10% of the height above the nominal ground level in front of the structure, with a minimum value of 0.5 m. For a supported wall, Δh shall be taken as 10% of the height beneath the lowest support, with a minimum value of 0.5 m.

The selection of the design water level shall take into account locally available data on the hydraulic and hydrogeological conditions at the site. The possibility of adverse water pressure conditions, such as those due to the presence of perched or artesian water tables or those due to saturation under heavy rainfall, shall be considered.

TABLE 13.3.1(A)
RANGE OF VALUES OF GEOTECHNICAL STRENGTH REDUCTION
FACTOR (ϕ_g) FOR RETAINING WALLS AND ABUTMENTS

Method of assessment of ultimate geotechnical strength	Range of values of ϕ_g		
	Bearing failure	Overturning, sliding and global stability	
		Permanent structures	Temporary structures
Analysis using geotechnical parameters based on appropriate advanced in situ tests	0.45–0.55	0.55	0.60–0.70
Analysis using geotechnical parameters from appropriate advanced laboratory tests	0.40–0.50	0.55	0.55–0.65
Analysis using CPT tests	0.35–0.45	0.50–0.55	0.50–0.60
Analysis using SPT tests	0.30–0.40	0.45–0.55	0.45–0.55

NOTE: Examples of testing regimes are given in AS 5100.3 Supp 1.

TABLE 13.3.1(B)
GUIDE FOR ASSESSMENT OF GEOTECHNICAL REDUCTION
FACTOR (ϕ_g) FOR RETAINING WALLS AND ABUTMENTS

Lower end of range	Upper end of range
Limited site investigation	Comprehensive site investigation
Simple methods of calculation	More sophisticated design method
Limited construction control	Rigorous construction control
Severe consequences of failure	Less severe consequences of failure
Significant cyclic loading	Mainly static loading
Use of published correlations for design parameters	Use of site-specific correlations for design parameters

13.3.2 Calculation of earth pressures

Calculation of the design action effects arising from earth pressures shall take into account the following factors:

- Surcharges on and slope of the ground surface.
- Inclination of the wall or structure face to the vertical.
- Water table levels, variations in these levels and seepage forces in the ground.
- Amount and direction of wall movement relative to the ground.
- Shear strength and unit weight of the ground.
- Rigidity of the wall and the supporting system.
- Wall roughness.
- Effects of any compaction during construction.
- Influence of surcharge loadings adjacent to the wall or abutment.
- Earthquake loads.

NOTES:

- Traditional methods of calculating active earth pressures (e.g., Rankine's method or Coulomb's method) may be employed.
- For passive earth pressure calculation, traditional methods such as Rankine's method and Coulomb's method are often unreliable. It is preferable to use a more rigorous method such as that described by Lee and Herington (1972), and Caquot and Kerisel (1948).

13.3.3 *Design for eccentric and inclined loads*

In assessing the ultimate geotechnical strength (R_{ug}), allowance shall be made for the possibility of very high edge stresses and a reduced effective contact area between the retaining wall or abutment footing and the ground as a result of load eccentricity.

13.3.4 *Design for serviceability*

For the serviceability design of retaining walls and abutments, the provisions of Clause 7.5 shall apply. In estimating the settlement and horizontal displacements, account shall be taken of the stiffness of the ground and the structural elements, and of the sequence of construction.

Allowable displacements for walls and abutments shall be established, taking into account the tolerance to deformation of the supported structures and services.

NOTES:

- 1 When no movement of the retaining structure relative to the ground takes place, the earth pressure may be calculated for the at-rest state of stress in the ground. This stress state will depend on the stress history of the ground. At-rest conditions can be expected to exist in the ground behind a retaining structure if the horizontal movement of the structure is less than about 0.05% of the unsupported height of the structure.
- 2 If a linear analysis is employed, the stiffnesses for the ground and structural elements should be appropriate for the level of deformation computed.

13.3.5 *Design for durability*

Design for durability shall be in accordance with Clause 9. The design life shall be in accordance with AS 5100.1.

Where materials other than concrete and steel are to be used for the construction of the structure, the requirements for durability in the relevant Standard for that material shall apply, unless otherwise specified by the relevant authority.

Where no Standard applies to the materials used in the structure, the requirements of the relevant authority shall apply.

13.4 **Structural design and detailing**

13.4.1 *General*

Structural design and detailing for retaining walls and abutments built of concrete and steel shall be in accordance with AS 5100.5 or AS 5100.6, as appropriate.

Where materials other than concrete and steel are to be used for the construction of the structure, then the requirements of the relevant Standard for that material shall apply to the structural design and detailing of the structure, unless otherwise specified by the relevant authority.

Where no Standard applies to the materials used for the construction of the structure, the requirements of the relevant authority shall apply.

Tensile stresses shall not be permitted in masonry and unreinforced concrete retaining walls and abutments.

13.4.2 *Joints*

Vertical contraction joints shall be provided in long concrete retaining walls and abutments to control indiscriminate shrinkage cracking. Where the structure is founded directly on rock, a reduced joint spacing shall be used.

NOTE: Contraction joints are recommended at a spacing of 8 m to 10 m along substructure members on other than rock. Where the structure is founded on rock, a reduced spacing of 5 m is recommended.

Where expansion joints are provided, suitable compressible jointing material shall be provided in the expansion joints.

NOTE: Expansion joints are recommended at a spacing of 30 m along substructure members.

Contraction or expansion joints shall also be provided where abrupt changes in structure section occurs. For counterfort walls, expansion joints may be provided either between double counterforts, or midway between counterforts.

Where there is the possibility of water seepage through joints, either a water-stop within the joint or a flexible waterproof membrane behind the joint shall be used.

Provision for shear transfer shall be made for all joints.

13.4.3 Shrinkage and temperature reinforcement

All reinforced concrete retaining walls and abutments shall be reinforced for shrinkage and temperature effects to the requirements of AS 5100.5.

13.5 Materials and construction requirements

Materials and construction requirements for retaining walls and abutments built of concrete and steel shall be in accordance with AS 5100.5 or AS 5100.6, as appropriate.

Where materials other than concrete and steel are to be used for the construction of the structure, then the requirements of the relevant Standard for that material shall apply, unless otherwise specified by the relevant authority.

Where no Standard applies to the materials used for the construction of the structure, the requirements of the relevant authority shall apply.

13.6 Drainage

Unless hydrostatic pressure is taken into account in design, effective drainage shall be provided behind retaining walls and abutments to permanently relieve water pressures. Where the safety and serviceability of the design depends on the successful performance of the drainage system, the consequences of failure of the drainage system shall be considered, and measures shall be taken to ensure continuing performance of the drainage system. Details of the drainage system shall be subject to the approval of the relevant authority.

NOTE: The seepage quantities, pressures, and chemical content of water emerging from a drainage system should be considered, and appropriate measures taken to dispose of this water.

14 BURIED STRUCTURES

14.1 Scope

The requirements for the design of structures where soil and rock loads form a significant proportion of the total loads on the structure shall be as set out herein unless approved otherwise by the relevant authority.

Precast concrete box culverts shall be designed in accordance with AS 1597, for the sizes specified in that Standard. For the design of sizes larger than those specified in AS 1597, the principles of that Standard shall apply.

NOTE: The design of buried arch structures is a specialized field and should be carried out by experienced design engineers.

14.2 Loads and load combinations

Buried structures shall be designed for the loads and other actions set out in Clause 8.2.

The load combinations for strength, stability and serviceability shall be as specified in Clauses 8.3.3 and 8.4.

The following additional loads and actions shall be considered when determining the design loads for buried structures:

- (a) Variations in soil density, stiffness, or strength across the structure, or through the depth of the soil over and around the structure.
- (b) The effects of structure stiffness on the interaction between the ground and the structure.
- (c) Transverse or longitudinal loads due to fill slopes or retaining walls above the structure, or construction on a slope.
- (d) Loads in precast elements occurring during handling and erection.
- (e) Varying load and restraint conditions during backfilling operations.
- (f) Locked-in stresses due to compaction loads and deflection of the structure during backfill.
- (g) Loads due to ground water, taking into account variations in the level of ground water.
- (h) Effects due to distortion of the structure.

The design shall take into account non-linear and non-elastic behaviour of the soil and the structure where these effects may be significant. Axial loads shall be considered.

14.3 Design requirements

14.3.1 Design for strength and stability

For the geotechnical and structural design of buried structures, the provisions of Clause 7.3.3 shall apply. In designing for stability of buried structures, the provisions of Clause 7.4 shall apply.

The design geotechnical strength and design structural strength shall be calculated as the appropriate ultimate strength (R_u) multiplied by the appropriate strength reduction factor (ϕ). The structural strength reduction factor (ϕ_s) shall be obtained from AS 5100.5 or AS 5100.6, as appropriate. The geotechnical strength reduction factor (ϕ_g) shall be selected in accordance with Clause 7.3.5, and Tables 14.3.1(A) and 14.3.1(B).

Consideration shall be given to the possibility of failure due to loss of overall stability. The stability of the structure in all directions, for all possible modes of failure, shall be considered.

NOTE: The longitudinal stability of segmental structures such as culverts passing under embankment slopes, or constructed on a steep longitudinal gradient, should be given particular attention.

Foundations of buried structures shall be designed in accordance with Clauses 10 and 11, where appropriate.

TABLE 14.3.1(A)
RANGE OF VALUES OF GEOTECHNICAL STRENGTH
REDUCTION FACTOR (ϕ_g) FOR BURIED STRUCTURES

Method of assessment of ultimate geotechnical strength	Range of values of ϕ_g
Analysis using geotechnical parameters based on appropriate advanced in situ tests	0.50–0.65
Analysis using geotechnical parameters from appropriate advanced laboratory tests	0.45–0.60
Analysis using CPT tests	0.40–0.50
Analysis using SPT tests	0.35–0.40

NOTE: Examples of testing regimes are given in AS 5100.3 Supp 1.

TABLE 14.3.1(B)
GUIDE FOR ASSESSMENT OF GEOTECHNICAL STRENGTH
REDUCTION FACTOR (ϕ_g) FOR BURIED STRUCTURES

Lower end of range	Upper end of range
Limited site investigation	Comprehensive site investigation
Simple methods of calculation	More sophisticated design method
Limited construction control	Rigorous construction control
Severe consequences of failure	Less severe consequences of failure
Significant cyclic loading	Mainly static loading
Use of published correlations for design parameters	Use of site-specific correlations for design parameters

14.3.2 *Design for serviceability*

For the serviceability design of buried structures, the provisions of Clause 7.5 shall apply.

14.3.3 *Design for durability*

Design for durability shall be in accordance with Clause 9. The design life shall be in accordance with AS 5100.1.

Where materials other than concrete and steel are to be used for the construction of the structure, the requirements for durability in the relevant Standard for that material shall apply, unless otherwise specified by the relevant authority.

Where no Standard applies to the materials used in the structure, then the requirements of the relevant authority shall apply.

14.4 Structural design and detailing

Structural design and detailing for buried structures built of concrete and steel shall be in accordance with AS 5100.5 or AS 5100.6, as appropriate.

Buried structures may be subject to high axial loads. Compression reinforcement for the design axial loads for concrete structures shall be designed in accordance with the requirements of AS 5100.5, where necessary, and shall meet the requirements of the relevant authority.

Where materials other than concrete and steel are to be used for the construction of the structure, then the requirements of the relevant Standard for that material shall apply to the structural design and detailing of the structure, unless otherwise specified by the relevant authority.

Where no Standard applies to the materials used for the construction of the structure, then the requirements of the relevant authority shall apply.

14.5 Materials and construction requirements

Materials and construction requirements for buried structures built of concrete and steel shall be in accordance with AS 5100.5 or AS 5100.6, as appropriate.

Where materials other than concrete and steel are to be used for the construction of the structure, then the requirements of the relevant Standard for that material shall apply, unless otherwise specified by the relevant authority.

Where no Standard applies to the materials used for the construction of the structure, then the requirements of the relevant authority shall apply.

APPENDIX A

ASSESSMENT OF GEOTECHNICAL STRENGTH REDUCTION FACTORS (ϕ_g)
FOR PILES

(Normative)

The geotechnical strength reduction factor (ϕ_g) shall be chosen, taking into account the factors that may influence the reliability of the ultimate geotechnical strength. A range of values is given in Table A1. Values of ϕ_g in excess of the given ranges shall only be used in exceptional circumstances backed by detailed quantitative justification.

In assessing the value to be chosen within the ranges specified, consideration shall be given to the factors given in Table A2, and appropriate judgement shall be exercised.

TABLE A1
RANGE OF VALUES FOR GEOTECHNICAL
STRENGTH REDUCTION FACTOR (ϕ_g)

Method of assessment of ultimate geotechnical strength	Range of values of ϕ_g
Static load testing to failure	0.70–0.90
Static proof (not to failure) load testing (see Note 1)	0.70–0.90
Dynamic load testing to failure supported by signal matching (see Note 2)	0.65–0.85
Dynamic load testing to failure not supported by signal matching	0.50–0.70
Dynamic proof (not to failure) load testing supported by signal matching (see Notes 1 and 2)	0.65–0.85
Dynamic proof (not to failure) load testing not supported by signal matching (see Note 1)	0.50–0.70
Static analysis using CPT data	0.45–0.65
Static analysis using SPT data in cohesionless soils	0.40–0.55
Static analysis using laboratory data for cohesive soils	0.45–0.55
Dynamic analysis using wave equation method	0.45–0.55
Dynamic analysis using driving equation for piles in rock	0.50–0.65
Dynamic analysis using driving equation for piles in sand	0.45–0.55
Dynamic analysis using driving equation for piles in clay	(see Note 3)
Measurement during installation of proprietary displacement piles, using well-established in-house equation	0.50–0.65

NOTES:

- 1 ϕ_g should be applied to the maximum load applied.
- 2 Signal matching of the recorded data obtained from dynamic load testing should be undertaken on representative test piles using a full wave signal matching process.
- 3 Caution should be exercised in the sole use of dynamic equation (e.g., Hiley) for the determination of the ultimate geotechnical strength of piles in clays. In particular, the dynamic measurements will not measure the set-up that occurs after completion of driving. It is preferable that assessment be first made by other methods, with correlation then made with dynamic methods on a site-specific basis if these latter are to be used for site driving control.
- 4 For cases not covered by Table A1, values of ϕ_g should be chosen using the stated values as a guide.

TABLE A2
ASSESSMENT OF GEOTECHNICAL
STRENGTH REDUCTION FACTOR (ϕ_g)

Circumstances in which lower end of range may be appropriate	Circumstances in which upper end of range may be appropriate
Limited site investigation	Comprehensive site investigation
Simple method of calculation	More sophisticated design method
Average geotechnical properties used	Geotechnical properties chosen conservatively
Use of published correlations for design parameters	Use of site-specific correlations for design parameters
Limited construction control	Careful construction control
Less than 3% piles dynamically tested	15% or more piles dynamically tested
Less than 1% piles statically tested	3% or more piles statically tested

APPENDIX B
ON-SITE ASSESSMENT TESTS OF ANCHORAGES
(Informative)

B1 GENERAL

On-site testing of anchorages is required by Clause 12.6. Typical generic requirements for such testing are outlined herein.

B2 DEFINITIONS AND NOMENCLATURE

For the purpose of this Appendix, the definitions below apply.

B2.1 Free length (L_{fr})

That length, in metres, of a tendon between the anchorage assembly and the bond length, or transition length, which does not transfer any tendon load to the surrounding rock, concrete or other material through which the anchor passes.

B2.2 Effective free length (L_{ef})

The apparent length, in metres, over which the tendon is assumed to extend elastically as determined by stressing tests. It is calculated from the load/elastic displacement data following testing, to indicate the length of tendon that is apparently fully decoupled from the surrounding grout.

B2.3 Bond length (L_v)

That length, in metres, at the end of a tendon within which provision is made for the load transfer to the surrounding rock.

B2.4 Design working load (T_D)

The long-term load, in kilonewtons, that is required in the tendon.

B2.5 Lock-off load (T_o)

The load, in kilonewtons, equal to the design working load plus an allowance for loss of prestress.

B2.6 Test load (T_p)

The maximum load, in kilonewtons, to which a tendon is subjected in the short term for proof load and acceptance tests.

B2.7 Minimum breaking load (T_u)

The minimum breaking load, in kilonewtons, of the tendon. This is calculated from the minimum strength of the component material as nominated by the supplier and verified by test.

B2.8 Initial load (T_A)

The initial load, in kilonewtons, selected for proof load and acceptance tests.

B2.9 Residual load (T_R)

The load, in kilonewtons, remaining in the tendon at any time after lock-off, usually measured by a lift-off test.

B2.10 Lift-off test

The test to determine the residual load in the tendon. Lift-off occurs when an applied load in excess of the residual load causes a very small but perceptible movement of the stressing head, nut or other locking device away from the anchor baseplate (usual range of movement 0.2–1.0 mm).

B2.11 Notation

The following symbols are used in this Appendix:

- A_t = the cross-sectional area of tendon, in millimetres square, as determined by testing
- E_t = the modulus of elasticity of steel tendon, in megapascals, as determined by testing
- L_{ef} = the effective free length
- L_{fr} = the free length
- L_v = the bond strength
- T = the anchor load, in kilonewtons
- T_A = the initial load
- T_D = the design working load
- T_o = the lock-off load
- T_p = the test load
- T_R = the residual load
- T_{RC} = the calculated residual load immediately after lock-off
- T_u = the minimum breaking load
- δL = the total extension of tendon relative to a datum, in millimetres
- δL_e = the elastic extension of tendon at each load stage, in millimetres
- δL_r = the calculated elastic extension of tendon under test load (T_p), in millimetres
- δL_{pl} = the plastic or non recoverable extension of tendon at each load stage, in millimetres

B3 STRESSING PROCEDURES AND ASSESSMENT OF PROOF LOAD TESTS

B3.1 General

The procedure and assessment described in Paragraphs B3.2 and B3.3 should be adopted for all anchors that are specified or directed to be subject to proof load tests.

B3.2 Stressing procedure

The procedure for stressing is as follows:

- (a) Select an initial load (T_A) so that $0.1T_p \leq T_A \leq 0.2 T_p$. Use $T_p = 0.8T_u$. Divide the range between T_A and T_p into 6 to 10 approximately equal steps of magnitude δT .
- (b) Establish a datum to measure $\delta L = \delta L_e + \delta L_{pl}$. The movement of this datum under the influence of anchor stressing should not exceed 0.5% of the calculated anchor extension (δL_r).

- (c) Carry out a program of cyclic loading and unloading with the load being increased from T_A in successive cycles by δT , $2\delta T$, $3\delta T$, etc. until the specified maximum load T_p is reached. After the peak load in each cycle is reached, take measurements of the load decrease with the deformation held constant for a time interval **Error! Objects cannot be created from editing field codes.**, where **Error! Objects cannot be created from editing field codes.** should be 5 min and n should initially be 1, but may subsequently be increased to 3 and then to 10, if the limiting values given in Table B1 are exceeded. Alternatively, the measurements of the deformation that increase with the load held constant can be taken for the same time intervals. After the above measurements have been taken for each cycle, reduce the load to T_A and record the extension (δL).
- (d) After the cycle for the test load (T_p) has been carried out, undertake a further cycle in the following manner:
- (i) Firstly, take the tendon load to T_p and then reduce to $0.3T_p$ in four equal increments.
 - (ii) Secondly, increase the load in three equal increments to the lock-off load (T_o). For each of these load points, record the extension measurements.
 - (iii) Finally, carry out lock-off. During lock-off, measure the draw-in of the wedges or cones (if any are used in the anchor head), and determine the residual load by lift-off test. In addition, determine the zero friction line and the calculated residual load immediately after lock-off (T_{RC}) in accordance with Paragraph B3.3.
- (e) After 48 h, determine the residual load again by lift-off test.
- (f) If the loss of residual load exceeds the limit given in Paragraph B3.3(d), determine the residual load again after a further period of 48 h. If the limit given in Paragraph B3.3(d) for the second 48 h period is exceeded, determine the residual load again after a final 48 h period. The three 48 h periods should be continuous.

TABLE B1

LIMITING VALUES OF EXTENSION INCREASE AND LOAD LOSS

Condition	Observation period	Limiting value within observation period	
		Extension increase (a)	Load loss (b)
(A)	0 to δt	Max. 2% of δL_r	Max. 2% of T_p
(B)	δt to $3\delta t$	Max. 1% of δL_r	Max. 1% of T_p
(C)	$3\delta t$ to $8\delta t$	Max. 1% of δL_r	Max. 1% of T_p

NOTES:

- 1 (a) refers to test procedure where the load is kept constant during the observation period.
- 2 (b) refers to test procedure where the deformation is kept constant during the observation period.
- 3 If condition (A) is not satisfied, increase the observation period to $3\delta t$ and test for compliance with condition (B). If condition (B) is not satisfied, increase the observation period to $10\delta t$ and test for compliance with condition (C).

B3.3 Assessment

The following conditions should be satisfied:

- (a) *Change of load or deformation* The change of load or deformation should not exceed the values given in Table B1.

- (b) *Effective free length* The effective free length (L_{ef}) should be between the following limits up to the maximum test load (T_p):

$$0.9L_{fr} \leq L_{ef} \leq (L_{fr} + 0.5L_v) \quad \dots \text{B3.3(1)}$$

where

$$L_{ef} = \frac{\delta L_e(x) A_t E_t}{T(x) - T_A} \times 10^{-6} \quad \dots \text{B3.3(2)}$$

(x) refers to any point on the loading curve

- (c) *Residual load* The residual load measured in the immediate lift-off test should not be less than $1.1T_D$ nor greater than $1.15T_D$.
- (d) *Loss of residual load* The loss of residual load in the 48 h period immediately following lock-off (see Paragraph B3.2(e)) should not exceed 4% of the initial residual load.

If the loss exceeds 4%, the test may be repeated for two further 48 h period (as described in Paragraph B3.2(f)), and the anchor should be acceptable provided the total loss does not exceed 6% after the second 48 h period, or 7% after the third 48 h period.

- (e) *Draw-in of wedges* The draw-in of the locking cones/wedges (if any are used in the anchor head) should be within the limits given by the manufacturer of the anchor system.

Use the last six points of the final cycle (see Paragraph B3.2(d)) to determine a zero friction line by the least squares method and determine also the calculated residual load immediately after lock-off (T_{RC}).

Determine the plastic extensions (δL_{pl}) from the load versus extension plots.

B4 STRESSING PROCEDURES AND ASSESSMENT OF ACCEPTANCE TESTS

B4.1 General

The procedure and assessment described in Paragraphs B4.2 and B4.3 should be adopted for all anchors for which an acceptance test is specified or directed.

B4.2 Stressing procedure

The procedure for stressing is as follows:

- (a) Select an initial load (T_A) so that $0.1T_p \leq T_A \leq 0.2T_p$. Use $T_p = 1.5T_D$. $T_p \leq 0.8T_u$.
- (b) Establish a datum to measure $\delta L = \delta L_e + \delta L_{pl}$. The movement of this datum under the influence of anchor stressing should not exceed 0.5% of the calculated anchor extension (δL_r).
- (c) Load the anchor up to the test load (T_p) and take measurements of the load decrease with the deformation held constant for a time interval $n\delta t$, where δt should be 5 min and n should be 1 initially, but may be increased subsequently to 3 and then to 10 if the limiting values given in Table B1 are exceeded. Alternatively, take measurements of the deformation increase, with the load held constant over the same time intervals.
- Reduce the load to T_A and record the extension (δL).
- (d) After the required measurements have been taken for the final test cycle, increase the load to T_o . Carry out lock-off and measure the residual load immediately by a lift-off test. Unload the anchor completely prior to stressing to T_o if desired.
- (e) After 48 h, determine the residual load again by lift-off test.

- (f) If the loss of residual load exceeds the limit given in Paragraph B4.3(d), determine the residual load again after a further period of 48 h. If the limit given in Paragraph B4.3(d) for the second 48 h period is exceeded, determine the residual load again after a final 48 h period. The three 48 h periods should be continuous.

B4.3 Assessment

An anchor may be accepted for use at a different (usually lower) working load, provided it is re-tested and satisfies the following criteria:

- (a) *Change of load or deformation* The change of load or deformation does not exceed the values given in Table B1.
- (b) *Effective free length* The effective free length (L_{ef}) should be between the following limits up to the maximum test load (T_p):

$$0.9L_{ef} \leq L_{ef} \leq (L_{ef} + 0.5L_v) \quad \dots B4.3(1)$$

where

$$L_{ef} = \frac{\delta L_e(x) A_t E_t}{T(x) - T_A} \times 10^{-6} \quad \dots B4.3(2)$$

(x) refers to any point on the loading curve

- (c) *Residual load* The residual load measured in the immediate lift-off test should not be less than $1.1T_D$ nor greater than $1.15T_D$.
- (d) *Loss of residual load* The anchor should be acceptable provided the loss of residual load in the 48 h period immediately following lock-off (see Paragraph B4.2(e)) does not exceed 4% of the initial residual load.

If the loss exceeds 4% repeat the test for two further 48 h periods (as described in Paragraph B4.2(f)). The anchor should be acceptable provided the total loss does not exceed 6% after the second 48 h period or 7% after the third 48 h period.

If an anchor does not satisfy Items (a), (b), (c) and (d), it should not be accepted.

Determine the plastic extensions (δL_{pl}) from the load versus extension plot and conform to that obtained in an appropriate proof load test.

AMENDMENT CONTROL SHEET

AS 5100.3—2004

Amendment No. 1 (2010)

CORRECTION

SUMMARY: This Amendment applies to Clause 12.3.5(c) (new).

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